



OPEN Hybrid deep models for parallel feature extraction and enhanced emotion state classification

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Emotions play a vital role in recognizing a person's thoughts and vary significantly with stress levels. Emotion and stress classification have gained considerable attention in robotics and artificial intelligence applications. While numerous methods based on machine learning techniques provide average classification performance, recent deep learning approaches offer enhanced results. This research presents a hybrid deep learning model that extracts features using AlexNet and DenseNet models, followed by feature fusion and dimensionality reduction via Principal Component Analysis (PCA). The reduced features are then classified using a multi-class Support Vector Machine (SVM) to categorize different types of emotions. The proposed model was evaluated using the DEAP and EEG Brainwave datasets, both well-suited for emotion analysis due to their comprehensive EEG signal recordings and diverse emotional stimuli. The DEAP dataset includes EEG signals from 32 participants who watched 40 one-minute music videos, while the EEG Brainwave dataset categorizes emotions into positive, negative, and neutral based on EEG recordings from participants exposed to six different film clips. The proposed model achieved an accuracy of 95.54% and 97.26% for valence and arousal categories in the DEAP dataset, respectively, and 98.42% for the EEG Brainwave dataset. These results significantly outperform existing methods, demonstrating the model's superior performance in terms of precision, recall, F1-score, specificity, and Mathew correlation coefficient. The integration of AlexNet and DenseNet, combined with PCA and multi-class SVM, makes this approach particularly effective for capturing the intricate patterns in EEG data, highlighting its potential for applications in human-computer interaction and mental health monitoring, marking a significant advancement over traditional methods.

Keywords Emotion analysis, Deep learning, AlexNet, DenseNet, Support vector machine, Principal component analysis

Human emotions are directly related to behavior. The behavioral and physiological response is reflected through emotions. In interpersonal communication, emotions play a vital role. Emotions determines the quality of daily life and its related needs. When the needs are met, humans will generate positive emotions and if it is not then negative emotions will be generated. In some cases, neutral emotions will be generated so that others could not be able to find the person's emotional status. Accurate emotion recognition is crucial in various applications, significantly enhancing their effectiveness and user experience. In mental health monitoring, it can detect signs of emotional disorders, enabling timely intervention. Human-Computer Interaction (HCI) benefits by allowing systems like virtual assistants to respond empathetically, improving user satisfaction. In education, emotion recognition personalizes learning, adjusting content based on student engagement and frustration levels. The gaming industry uses it to create adaptive and immersive experiences, while market research leverages it to analyze consumer responses for better marketing strategies. Additionally, assistive technologies for individuals with disabilities, such as autism, can help users navigate social interactions more effectively by providing real-time emotional feedback. These use cases highlight the importance of accurate emotion recognition in enhancing functionality and user engagement across diverse fields.

In recent times, the study of human computer interactions gained more attention and researchers put more efforts into developing a natural interaction model so that the computers can understand the emotional statuses of the user. For better emotion classification, body gestures, speech, facial expressions are used by the research

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community. However, these modalities can be artificially generated. For example, if a person is angry, then they can even smile to hide their anger emotion. Meanwhile other factors like poor lighting, noise artifacts and usage of accessories like glasses reduces the classification performances. Thus, to attain a reliable and more efficient emotion classification, physiological data has been used in recent times. The physiological data included electroencephalography (EEG) data which can be collected from a person directly. EEG data collects the brain electrical activity, and it provides good temporal resolution, so that emotion recognition applications can understand the brain activity and define the emotion states of a person.

Earlier machine learning models were used to EEG data classification which requires additional feature processing modules. Moreover, the results of machine learning models are not accurate, and its feature processing abilities are limited. Recently deep learning models are employed over machine learning techniques^{1–4}. Specifically, convolutional neural network is widely used in various application. Emotion classification using convolutional neural network was employed in our earlier research work and attained better performances than machine learning models. However, for enhanced performance and to select the best features in the emotion classification process, in this research work a hybrid deep learning model is presented which incorporated convolutional neural network variants AlexNet and DenseNet algorithms.

Objectives and motivations

Despite significant advancements in emotion classification using machine learning and deep learning techniques, several gaps persist in the existing literature. Traditional machine learning methods often struggle with feature extraction and require extensive preprocessing, which can limit their accuracy and generalizability. Recent deep learning approaches, while more effective, typically focus on single-model architectures that may not fully capture the complex features present in physiological data such as EEG signals. Moreover, many existing models do not adequately address the issue of dimensionality reduction, leading to increased computational costs and potential overfitting. This research aims to bridge these gaps by introducing a hybrid deep learning model that utilizes the strengths of AlexNet and DenseNet for feature extraction, followed by dimensionality reduction using Principal Component Analysis (PCA) and classification with a multi-class Support Vector Machine (SVM). The primary motivation is to enhance the accuracy, efficiency, and robustness of emotion classification systems by addressing the limitations of current methodologies.

Research questions

This research work seeks to answer several key research questions to advance the field of emotion classification from physiological data. First, how can the integration of multiple deep learning models improve the feature extraction process for emotion classification? Second, what is the impact of dimensionality reduction on the performance and computational efficiency of hybrid deep learning models? Third, how does the proposed hybrid model compare to existing machine learning and deep learning approaches in terms of classification accuracy, precision, recall, F1-score, specificity, and Mathew correlation coefficient? By addressing these questions, this research aims to provide a comprehensive evaluation of the proposed model's effectiveness and its potential contributions to the field of emotion recognition.

The contributions made in this research work is presented as follows.

- Presented a hybrid deep learning models for extracting features from emotional data. Convolutional neural network variant, AlexNet and DenseNet are used for initial feature extraction. The extracted features are fused, and dimensionality is reduced using principal component analysis.
- Presented a classification model to classify the fused features from the deep learning models. A multi-class support vector machine is employed for feature classification.
- Presented experimental analysis using benchmark emotion datasets DEAP and EEG Brainwave dataset to validate the performances.
- Presented a detailed comparative analysis with existing emotion classification model to validate the proposed model better performances in terms of recall, precision, f1-score, specificity, Mathew correlation coefficient and accuracy.

The remaining discussions in the article are arranged in the following order. Section “[Related works](#)” presents a detailed literature review. Section “[Materials and methods](#)” presents the materials and methods. Details about the deep learning models and classification model are presented in this section. Section “[Results and discussion](#)” presents the experimental results and discussion; the summary is presented in the last section.

Related works

A detailed analysis of existing research work on emotion classification is presented. The existing methodologies and its features are studied in detail. The classification model reported in⁵ includes a machine learning algorithm to categorize seven discrete emotions. Based on the features extracted from the EEG signals, the emotions are categorized. Familiar machine learning model support vector machine is employed for classification. The target user is allowed to watch standardized videos and their emotions are obtained through EEG channels. Further the features from the EEG channel are extracted and classified through the support vector machine. The presented approach attained accuracy of 80.56% for positive emotions and 81.08% for negative emotions which is much lesser than the current approaches. The multimodal EEG signal-based emotion classification model presented in⁶ utilizes portable devices to capture the EEG signals. The captured signals are classified using supervised support vector machine. Compared to other machine learning based emotion classification models, the performance of the presented multi-modal approach is much better in terms of accuracy and deployment cost. However, the attained accuracy might vary with other approaches as the low-cost devices are prone to environmental noises.

The emotion recognition model presented in⁷ incorporated machine learning algorithms for feature classification from EEG signals. The presented approach employed machine learning for feature extraction and classification. Further an activation curve is obtained considering the emotion coefficients and classification results. The emotions are classified, and the emotional activation factors are revealed based on the obtained activation curve. In the last stage, a weight coefficient factor is used to improve the recognition accuracy. The emotion analyzes model presented in⁸ focused on the data imbalance issue in emotion classification process. The difference in the data distribution is mainly considered in the research work and introduces adversarial multitask training to establish better relationship between training and testing performances. The presented approach learns the common attributes and employs a gradient reversal layer to represent the source and target data classes. Due to this, the data imbalance issue is reduced, and the performance is increased in the classification process.

The emotion recognition model presented in⁹ includes discrete Fourier transform for kernel function to recognize the user emotions. Linear kernel and polynomial kernel functions are employed to handle the kernelized features and spatial features in the video frames. Finally, based on the features the emotions are classified with better accuracy compared to traditional methods. A spiking neural network model was reported in¹⁰ for recognizing emotions. The presented approach includes algorithms like discrete wavelet transform, fast Fourier transform, and variance Fourier transform for feature extraction. Further the extracted features are classified using spiking neural network and attained improved classification performance over conventional techniques.

In general, compared to traditional ML models, extreme learning models exhibit better performances^{11,12}. The machine learning model extreme gradient boosting based emotion classification approach was presented in¹³ to compute the correlation between emotional labels and EEG features. The presented multidomain model extracts the features using sliding time window and classify them using gradient boosting algorithm. Compared to single channel classification, the performance of multichannel is higher and it validated by experimentation using benchmark datasets. An automatic emotion recognition model presented in¹⁴ extracts the physiological time and frequency domain features to find the relevant emotions. The presented approach includes a forward floating kernel-based feature selection algorithm for optimal feature selection. Further, the selected features dimensions are reduced through generalized discriminant analysis model. Finally, the features are classified into four types of emotions using the least square support vector machine. However, the attained classification accuracy is comparatively lower than the current approaches.

The multichannel EEG emotion recognition model presented in¹⁵ dynamically learns the features using dynamic graph convolutional neural network model. The presented approach performs discriminative feature extraction which improves emotion recognition performances. Experimental results confirm the presented approach superiority in terms of accuracy over traditional machine learning based recognition models. Similar multi-channel emotion analysis model presented in¹⁶ detects emotions through emotion detection architecture. The presented architecture includes a specified feature extraction model and an emotion correlation learner. Each channel in the proposed architecture performs individual feature extraction and specifies the emotions as a hierarchical structure. Finally, the correlation between the emotions is learned to label the feature as in to positive, negative, and neutral categories.

The emotion recognition model presented in¹⁷ includes a bidirectional principal component analysis and least square linear discriminant analysis to improve the recognition efficiency. The presented approach initially extracts visual features using optimized kernel Laplacian radial basis function based neural network classifier. Further the features are classified based on a set of rules to emotion categorization. The emotion classification model reported in¹⁸ extracts the essential features from EEG signal using robust principal component analysis methods. The presented PCA model is embedded with transfer learning process to obtain the labeled data. Based on the intra and inter individual differences the irrelevant features are removed, and optimal features are selected in the presented model. Compared to traditional methods, the obtained accuracy for emotion category is less. A Bi-stream emotion attribution and classification network reported for emotion recognition, emotion summarization and emotion attribution tasks¹⁹. The presented classification network has an attribution model and a classification model in which the essential emotional segments are extracted by the attribution model, and it is classified by the classification model. The classification model is a bi-stream architecture which extracts the segments and provides classification results based on the ground truth labels.

The emotion recognition model presented in²⁰ includes deep learning algorithm LSTM and optimization algorithm to attain enhanced recognition performances. The presented approach utilizes a differential evolution model to optimize the network hyper parameters of LSTM model. Further comparative analysis with existing optimization models like particle swarm optimization, random search, simulated annealing models validates the better performance of the presented optimized LSTM model. The accuracy attained by the optimized LSTM is 77.68% which is much lesser than other approaches. The emotion classification model presented in²¹ employed machine learning algorithms like attention LSTM, k-nearest neighbor, and SVM to categorize six different types of emotions. The presented approach extracts the features from EEG, skin responses, respiration and ECG channels and categorize them into different types. Compared machine learning models kNN and SVM the performance of attention LSTM is better, and it validated through the experimental analysis.

The hybrid deep learning model presented in²² combines convolutional neural network with bidirectional LSTM to learn the time series emotion data. The presented model includes linear discriminant analysis to learn the low dimensional source data and target data. Further the dimensions are reduced using maximum mean discrepancy. The difference constraints between target and source data are handled through graph embedding procedure. Finally, a deep neural network is employed to classify the discriminative emotion features into different categories. Experimentations confirmed the deep learning model performance over machine learning approaches.

The emotion recognition model presented in²³ preserves the emotional information and neglects the irrelevant information to avoid errors in the recognition process. The presented recognition model includes an attention mechanism based convolutional recurrent neural network to learn the relevant features. Experimentations confirmed that the presented network effectively extracts the features and improves the recognition performances over conventional approaches. The emotion recognition model presented in²⁴ includes deep CNN and AlexNet models to learn the low- and high-level features from the input signal. The input is initially segmented into multiple portions and then the features are extracted using the deep learning models. The extracted low level and high-level features are combined through pyramid matching technique. Finally, machine learning classifiers are employed to classify the emotion states better than the existing methods. Similar deep learning model presented in^{25,26} includes CNN and 3DCNN model to handle high dimension features. The presented approach extracts the features using deep learning models and combine them using a fusion network to attain improved recognition performances.

An emotion classification model presented in²⁷ extracts the features like frequency bands, differential asymmetry, rational asymmetry, and differential entropy features. Further wrapper and filter method are combined to select the essential features. Finally, the features are classified using hybrid deep learning which combines random forest with deep learning model. In research work²⁸ multiple deep learning models like VGG16, SqueezeNet, DenseNet and ResNet variants are used to categorize different types of emotions. Based on the deep features and acoustic features, a hybrid feature vector is initially created. From the feature vector, Relief algorithm extracts the essential optimal feature. Finally, machine learning models are used to classify the selected features which categorizes different types of emotions. However, the attained accuracy is comparatively lower than the recent models and the computation complexity is high due to multiple feature processing elements. Table 1 provides the comparison of existing methods.

Challenges

- Many studies focus on addressing the issue of data imbalance in emotion classification. For instance, the approach in⁸ employs adversarial multitask training to handle data distribution differences, but this remains a common challenge across multiple works.
- The use of low-cost portable devices for capturing EEG signals, as mentioned in⁶, can lead to variability in accuracy due to environmental noises. This presents a challenge in ensuring consistent and reliable emotion classification results.
- Various methodologies employ different techniques for feature extraction and selection, such as PCA, generalized discriminant analysis, and wrapper-filter methods^{7,13,27}). The challenge lies in optimizing these techniques to improve recognition accuracy while maintaining computational efficiency.

Reference number	Techniques used	Evaluation metrics with numerical values	Merits	Limitations
29	InceptionV3, transfer learning, data augmentation	Accuracy: 98%, high precision, recall, F1-score, ROC-AUC	High accuracy, aids early detection, robust performance	Reliance on small dataset, need for validation with larger datasets
30	VGG-19, Inception-v3, XceptionNet, transfer learning	VGG-19: CK + 99%, FER2013 65%, JAFFE 94%; Inception-v3: CK + 99%, FER2013 86%, JAFFE 86%; XceptionNet: CK + 99%, FER2013 77%	Effective use of transfer learning, adaptable to varying data	Varying performance depending on dataset, need for refinement with larger datasets
31	InceptionV3, CNN	CK + 99.5%, JAFFE 86%, FER2013 73%	High accuracy, evaluates emotional patterns and mental health	Performance varies by dataset, optimization needed for diverse data
32	CNN (Convolutional Neural Network)	FER-2013: 79%, CK+: 95%, high precision, recall, F1-score, AUC	High accuracy, robust performance across datasets	Varying performance with different datasets, challenge with larger datasets
33	CNN, Gated Recurrent Units (GRU), SMOTE	Accuracy: 95.18%, high precision, recall, RMSE	High accuracy, improved energy efficiency, handles imbalanced data	Increased complexity and computational cost, need for optimization
34	Attention mechanism, transfer learning, Word2Vec, BERT	SST-2: 94.48%, SST-5: 56.23%, high precision, recall, F1-score	Improved classification performance, enhanced generalizability	Complexity of training, dependency on dataset similarity
35	CNN, GRU, data normalization, feature transformation	Accuracy: 98%, low RMSE, optimized with Adam optimizer	High accuracy, generalization due to dropout technique, scalability	Computational complexity, need for substantial resources
36	Conditional Wasserstein GAN (CWGAN), EEG signal preprocessing	Classification accuracy: Valence + 6.5%, Arousal + 3%, high precision, recall	Improved classification accuracy, robustness with augmentation	Complexity of training, dependency on quality of augmented data
37	Topic-based models, ontology-based models, semantic relatedness	Superior performance in semantic relatedness of geographic phrases	Fewer restrictions in real world, highlights model equivalences	Complex architecture, need for comprehensive understanding
38	Local Cluster Analysis (LCA), k-means, PDDP	Initial precision: 0.2766, LCA precision: 0.3957	High precision retrieval, simple, independent of system architecture	High clustering overhead, inconsistent query improvement
39	DCNN, G-HHO, Otsu thresholding, feature extraction (size, variance, mean), data augmentation	Accuracy: 0.97, Precision: 0.99, Recall: 0.95, F-measure: 0.96	High accuracy, precision, and recall, efficient in execution time and memory usage, robust performance with augmented MRI datasets	Need for further validation with real-world applications, potential for hybridization with other swarm algorithms
40	BERT model, TextBlob for sentiment labeling, BoW, TF-IDF, Word2Vec for feature extraction, various ML models (LR, RF, SVM, etc.), deep learning models (LSTM, GRU, CNN, etc.)	Accuracy: 99.3% (BERT), 97% (LR, LSTM) Precision: 99.352% (BERT) Recall: 99.350% (BERT) F1 Score: 99.351% (BERT)	Effective preprocessing improves model performance, Comprehensive evaluation across multiple models and datasets	- High computational cost and time complexity for training BERT and Limited to Twitter data, which may not be representative of all social media platforms

Table 1. Existing methods of emotion classification.

- (d) The use of advanced models like deep CNNs, LSTMs, and hybrid models^{20,22,24}) often results in high computational costs and complexity. Balancing model performance with computational efficiency is a persistent challenge in emotion classification research.
- (e) Achieving high accuracy and robustness in emotion classification is a critical challenge. Some approaches, like the optimized LSTM model in²⁰ and the PCA with transfer learning in¹⁸, report lower accuracies compared to current state-of-the-art methods, highlighting the difficulty in consistently attaining high-performance levels across different datasets and methodologies.

Based on the study it can be confirmed that the performance of deep learning models is better than the traditional emotion recognition and classification models. Hybrid approaches for emotion classification can attain improved performance over traditional methods. Thus, in this research work a hybrid deep learning model-based emotion classification is presented in the following section.

Materials and methods

Dataset

In this research work, two benchmark datasets are used for experimentation. The first dataset used in the research work is DEAP dataset⁴¹. The DEAP (Dataset for Emotion Analysis using Physiological signals) dataset is a widely used benchmark for emotion classification studies. It consists of data collected from 32 participants who watched 40 one-minute-long music videos. Each participant's physiological responses, including EEG signals, were recorded at a sampling rate of 512 Hz. The dataset includes both raw and preprocessed signals, with the latter downsampled to 128 Hz to reduce computational load. Each recording contains 40 trials per participant, resulting in a total of 1,280 trials (32 participants x 40 trials). The data is segmented into 60-second epochs, and the labels for each trial indicate the participants' self-reported levels of arousal, valence, and dominance on a scale from 1 to 9.

The EEG Brainwave dataset, sourced from Kaggle, is designed to classify emotions into three categories: positive, negative, and neutral. This dataset comprises EEG recordings from multiple participants exposed to six different film clips, each lasting approximately 2 min. The EEG signals were recorded at a sampling rate of 150 Hz. Data collection was spread over several days to minimize participant fatigue and ensure the reliability of the recordings. The dataset includes a total of 324,000 data points, segmented into 3-minute epochs to capture the variations in brainwave patterns corresponding to different emotional states.

Data preprocessing

The proposed model data processing step includes epoching, standardization, min-max scaling, and z-score normalization techniques. Initially in the epoching process, from the continuous EEG signal, a specific time window is extracted. Further a standardization procedure is employed to avoid data overfitting. Two methods are employed in the standardization such as min-max scaling and z-score normalization. Min-Max scaling initially scales the data into a fixed range from 0 to 1. The mathematical formulation for the min-max scaling is given as

$$x_{norm} = \frac{x - x_{min}}{x_{max} - x_{min}} \quad (1)$$

where minimum range is indicated as x_{min} and maximum range is indicated as x_{max} . Next to min-max scaling, the data is normalized using Z-score normalization. In the z-score normalization, the features are rescaled so that the features will have standard normal distribution properties. z-score normalization is mathematically expressed as

$$z = \frac{x - \mu}{\sigma} \quad (2)$$

where μ indicates the mean and the σ indicates the standard deviation. Further from the normalized data, the features are extracted using deep learning algorithms like AlexNet, DenseNet 121 models.

AlexNet

The first feature extraction model used in the research work is a familiar convolutional neural network variant AlexNet. The architecture of AlexNet includes convolutional, pooling and fully connected layers. Initially the nonlinearity in the input is removed by applying an activation function. Rectified Linear Unit (ReLU) is widely used as activation function. In the initial convolution layer, the features are selected and fed into pooling layer. In the pooling layer, the features are down sampled to extract the rich features. Finally using the fully connected layer, the features are flattened so that the essential features are selected, and others are dropped. The general architecture of AlexNet includes a SoftMax function in the last layer for feature classification. However, in the proposed work, AlexNet is used for feature extraction thus it is removed from the architecture. The complete architecture of AlexNet and layer details are provided in Fig. 1.

The mathematical model for the convolution operation in AlexNet is formulated considering the feature map (I), bias factors (b_z) and convolutional kernel (k_z) as follows.

$$f_z = I * k_z + b_z \quad (3)$$

where f_z indicates the feature map which used in the convolution process. The operator $*$ indicates the convolution operation. As already mentioned, AlexNet utilizes non-linear activation function to introduce better non-linearity in feature map output. Mathematically the process is formulated as

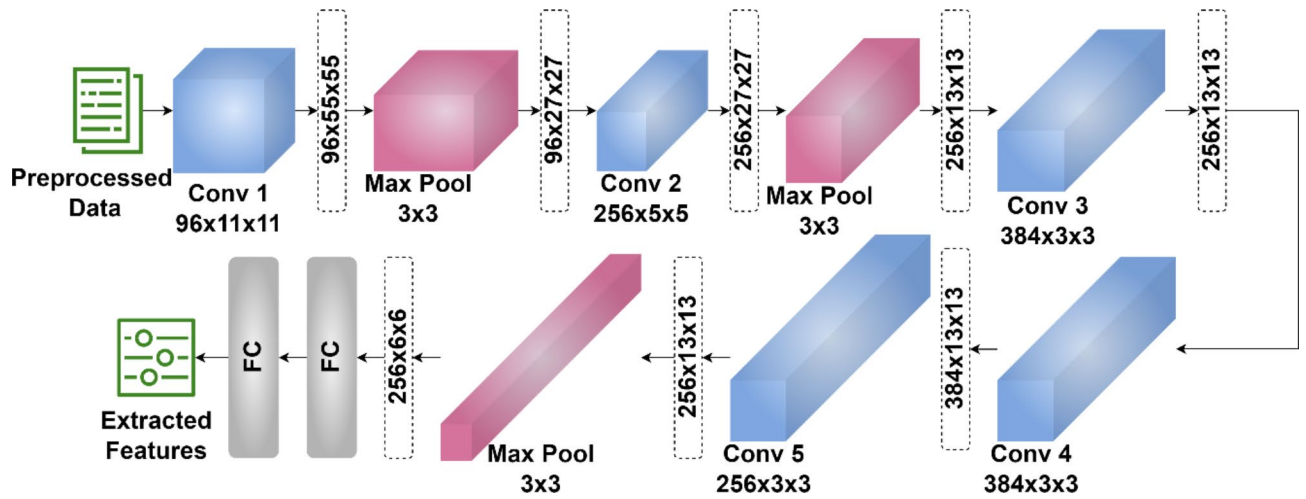


Fig. 1. Feature extraction using AlexNet.

$$A_{x,y} = a \left(\sum_{d=0}^{D-1} \sum_{p=0}^{s-1} \sum_{q=0}^{s-1} w_{p,q,d} I_{x+p,y+q,d} + b \right) \quad (4)$$

where feature map location is represented as x, y , input feature pixel value, weight and convolution kernel are represented using a function $I_{x+p,y+q,d}$.

Generally pooling layers in the deep learning algorithms are used to reduce the feature dimensions. AlexNet architecture includes two pooling layers after the convolution layer, and the pooling operation is mathematically expressed as

$$N_{x,y,z} = PL_{(p,q) \in R_{x,y}} (I_{p,q,z}) \quad (5)$$

where $R_{x,y}$ represents the pooling region, Pooling function is represented through PL . Max pooling is used in the proposed architecture, so that the largest convolved feature from the convolution output is selected for the pooling function. Based on the selected feature, the next feature output is provided to the next layer. This repeated operation in the AlexNet selects the optimal features from the data.

DenseNet 121

In the proposed model, in addition to AlexNet, the features are extracted through DenseNet algorithm. Each deep learning model has unique feature handling characteristics, thus in addition to AlexNet, another variant of CNN, DenseNet model is employed in the proposed work for feature extraction. The architecture of DenseNet is similar to traditional CNN with dense blocks. The feed forward deep learning architecture handles the local and global information from the input. Convolution, pooling, fully connected layer, and activation functions are major elements of DenseNet architecture. In the proposed work DenseNet 121 model is used. The convolution operation in the DenseNet is mathematically expressed using the convolution layer weights $(w_{k,j}^i)$ and bias factors (b_k^i) as follows

$$c_k^i = \left\{ b_k^i + \sum_{j=1}^{n^{i-1}} w_{k,j}^i \times x_j^{i-1} \right\} \quad (6)$$

where x_j^{i-1} indicates the input, convolutional layer output is represented as c_k^i . The pooling layer in the DenseNet is mathematically formulated as

$$c_k^i(max) = Pool_{max}(ReLU(x)), ReLU(x) \in c_k^i \quad (7)$$

where $ReLU$ represents the nonlinear activation function Rectified Linear unit which is mathematically expressed as

$$ReLU(x) = \max(0, x) \quad (8)$$

Finally, the fully connected layer flattens the features into one-dimensional features vectors. In the last layer of traditional DenseNet a classifier function is used to classify the features. Generally, SoftMax is used as a classifier function. In the proposed work, the features are extracted using DenseNet thus, the architecture neglected the final classifier function. The dense block in the architecture transfers the features from one block to another block to avoid errors and to reduce the computation cost. The essential formations that describe the dense layer function is given as follows.

$$s_n = z_n(s_{n-1}) \quad (9)$$

$$\mathcal{S}_n = \mathcal{Z}_n(s_{n-1}) + s_{n-1} \quad (10)$$

$$\mathcal{S}_n = \mathcal{Z}_n([s_0, s_1, s_2, \dots, s_{m-1}]) \quad (11)$$

where the nonlinear operation is represented as z , layer index is represented as m and m^{th} layer feature is represented as s_m . Figure 2 depicts a detailed illustration of feature extraction process of DenseNet model.

AlexNet and DenseNet were specifically chosen for feature extraction in our hybrid deep learning model due to their proven effectiveness in handling complex image data and their complementary strengths in capturing different aspects of features. AlexNet is known for its role in the field of deep learning, particularly in image recognition tasks. Its architecture includes multiple convolutional layers followed by max-pooling layers, which are highly effective at capturing spatial hierarchies in the input data. The use of ReLU activation functions accelerates the training process by introducing non-linearity, which is crucial for learning complex patterns in EEG signals. AlexNet's relatively simple yet powerful architecture allows for efficient feature extraction with a reduced risk of overfitting, making it suitable for emotion recognition where the nuances of EEG data need to be captured effectively.

DenseNet, on the other hand, introduces dense connections between layers, ensuring maximum information flow and gradient propagation throughout the network. This architecture mitigates the vanishing gradient problem and allows for deeper networks without the risk of information loss. Each layer in DenseNet receives inputs from all preceding layers, which promotes feature reuse and leads to more robust and discriminative feature representations. This is particularly beneficial for emotion recognition tasks, where the ability to capture intricate and subtle patterns in the EEG data is crucial.

The combination of AlexNet and DenseNet utilizes the strengths of both architectures. AlexNet provides a strong baseline for capturing basic spatial features, while DenseNet enhances this by capturing more complex and deeper feature interactions. This complementary approach ensures a comprehensive feature extraction process, capturing both high-level abstractions and detailed patterns in the EEG data. By fusing the features extracted from both models, our hybrid approach benefits from the strengths of both architectures, leading to more accurate and robust emotion recognition.

In summary, AlexNet and DenseNet were chosen for their proven capabilities in image recognition, their complementary strengths in feature extraction, and their suitability for capturing the complex patterns inherent in EEG signals necessary for accurate emotion recognition.

Feature fusion and dimensionality reduction

The features extracted from the AlexNet and DenseNet models are fused, and the high dimensional features are further converted into low dimensional features using principle component analysis. The simple feature fusion process concatenates both the features obtained from the AlexNet and DenseNet models. When combining both model features, the dimensionality of the feature is increased and also the feature map has redundant data as both models select the essential features from the input. To overcome this dimensionality issue, Principal component analysis (PCA) is employed in the proposed model. PCA is widely used whenever redundancy in variables or features maps. Principal Component Analysis (PCA) was selected for dimensionality reduction due to its effectiveness, simplicity, and interpretability in handling high-dimensional EEG data. PCA's linear nature makes it computationally efficient and easier to understand compared to non-linear techniques like t-SNE and autoencoders. t-SNE, while useful for visualization, is computationally intensive and less practical for feature extraction, and autoencoders require complex architectures and extensive tuning. PCA preserves the global structure of the data by maximizing variance, ensuring that the most informative features are retained. To determine the number of principal components to retain, we conducted an explained variance analysis, aiming to capture at least 95% of the total variance. This balance ensures that the dimensionality is reduced effectively without significant loss of important information, enhancing the overall performance of the emotion classification model.

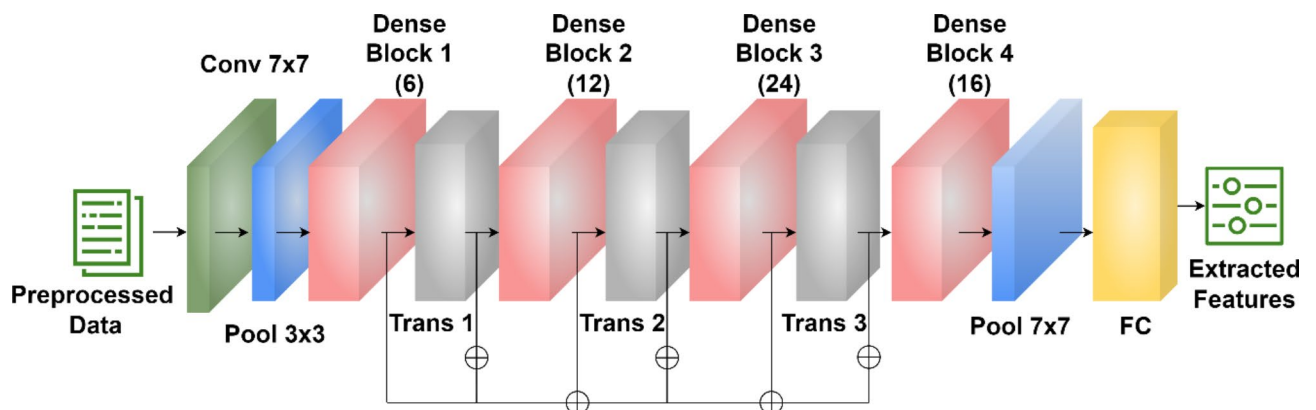


Fig. 2. DenseNet 201 architecture.

The term redundancy in the proposed work indicates the correlated features. For examples, if a feature is selected by the AlexNet model, the same feature will be selected by DenseNet model if the feature is highly important. In that case when concatenating the features, the selected feature will be repeated. In another case, if the features are correlated with one another, then it will increase the computation cost. Thus, PCA is used in the proposed work. PCA is familiar for data compression, and it can transform original multivariate data into linearly dependent variable datasets. The function which describes the PCA is given as follows.

$$(u, v) = \min_{u, v} \|x - uv\|^2 = \sum_{i=1}^n (x_i - uv_i)^2 \quad (12)$$

where the initial projection vectors are represented as u which is given as $u = (u_1, u_2, \dots, u_k)$ and the data after projection is represented as v which is given as $v = (v_1, v_2, \dots, v_n)$. Further the extracted dimensional reduced features are classified using multi-class support vector machine to categorize the emotion states.

Proposed emotion classification model

The proposed classification model depicted in Fig. 3 initially preprocesses the data. Preprocessing operations like epoching, standardization and normalization are performed. Further the preprocessed data is fed into deep learning models like Alexnet and DenseNet for optimal feature extraction. The extracted features are further fused, and the dimensionality is reduced using principal component analysis. The dimensionality reduced features are finally classified using multi class support vector machine.

The final classifier used in the proposed model for feature classification is multiclass support vector machine (MSVM). Traditional SVM is a binary classifier and in the proposed model different types of emotions must be categorized. Thus, multiclass SVM is employed. Support vector machine classifies the features through hyperplane. The kernel function used in the classifier process the feature map into feature space. The proposed model, multiclass classifier function is mathematically expressed as.

$$classifier = \frac{n(n-1)}{2} \quad (13)$$

where n represents the class, and its mathematical expression is given as follows.

$$n = \min_{w^{ij}, b^{ij}, \xi} \frac{1}{2} (w^{ij})^T (w^{ij}) + C \sum_t \xi_t^{ij} \quad (14)$$

where,

$$(w^{ij})^T \varphi \{x(t)\} + b^{ij} \geq 1 - \xi_t^{ij} \text{ if } y_t = i \quad (15)$$

$$(w^{ij})^T \varphi \{x(t)\} + b^{ij} \leq -1 + \xi_t^{ij} \text{ if } y_t = j \quad (16)$$

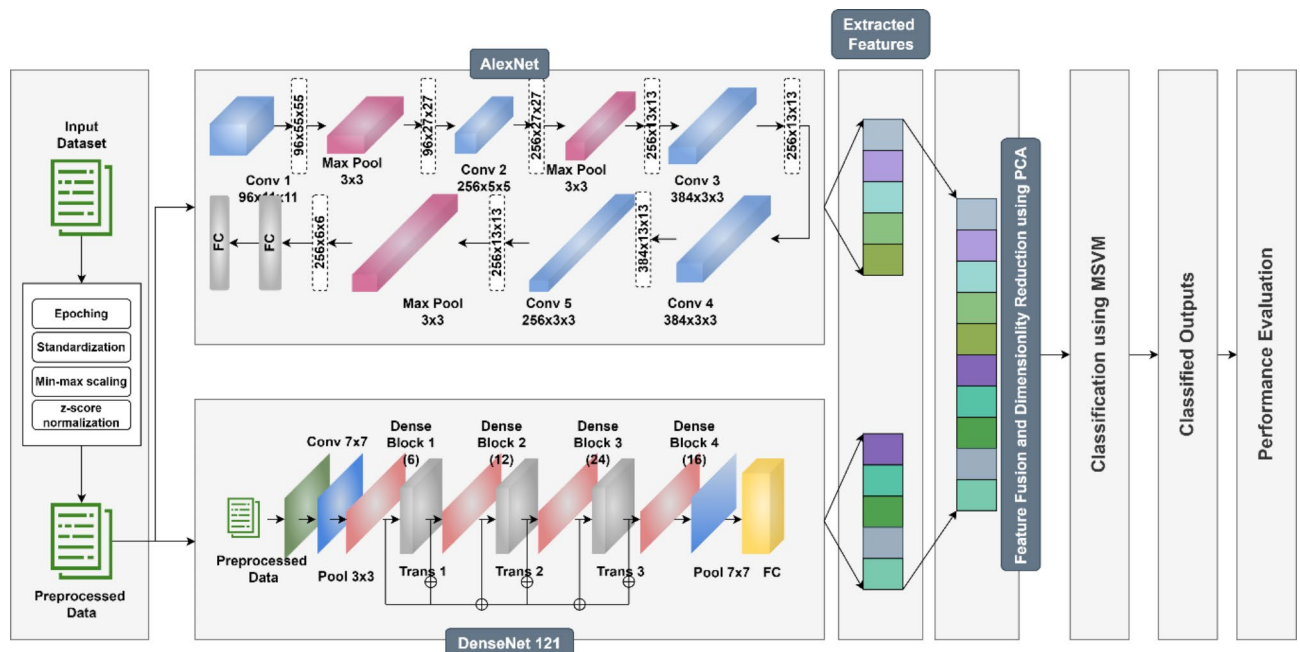


Fig. 3. Proposed hybrid deep learning model for emotion classification.

$$\xi_t^{ij} \geq 0$$

(17)

where ξ_t^{ij} represents the distance factor which is used to correct the margin of the hyperplane. C represents the regularization parameter and $\varphi \{x(t)\}$ represents the transformed input space vector. $(w^{ij})^T (w^{ij})$ represents the normal vector product and b^{ij} represents the bias factors. The feature space effectively categorizes the different types of emotions and improves the overall classification performances.

In our research, embedding techniques play a crucial role in transforming high-dimensional data into a lower-dimensional space that preserves essential information for emotion classification. Specifically, we employ Principal Component Analysis (PCA) as our primary embedding technique to achieve this transformation.

Feature extraction and fusion

Initially, features are extracted from the EEG data using two different deep learning models, AlexNet and DenseNet. These models were selected due to their proven efficiency in handling complex image data, which is analogous to the multi-dimensional nature of EEG signals. The extracted features from both models are then concatenated to form a comprehensive feature set that captures diverse aspects of the input data. However, this concatenation leads to an increase in the dimensionality of the feature space, which can introduce redundancy and computational challenges.

Dimensionality reduction using PCA

To address the issue of high dimensionality, we employ Principal Component Analysis (PCA), a widely used technique for reducing the dimensionality of data while retaining its most important features. PCA works by identifying the principal components, which are the directions of maximum variance in the data. These components are orthogonal to each other and provide a new set of axes that capture the most significant variations in the data. By projecting the original high-dimensional features onto these principal components, PCA effectively reduces the number of dimensions while preserving the essential information required for classification.

Application of PCA in our model

In our model, after the feature extraction and fusion stage, the combined feature set undergoes PCA. The steps involved are as follows.

- Compute the Mean: Calculate the mean of the combined feature set.
- Covariance Matrix: Compute the covariance matrix of the centered feature set (i.e., after subtracting the mean).
- Eigenvalues and Eigenvectors: Calculate the eigenvalues and eigenvectors of the covariance matrix. The eigenvectors corresponding to the largest eigenvalues represent the principal components.
- Projection: Project the original feature set onto the principal components to obtain the reduced-dimensional feature set.

This reduced-dimensional feature set is then fed into the multi-class Support Vector Machine (SVM) for final classification, enabling accurate and efficient emotion state recognition. Through this embedding technique, we enhance the overall performance of our hybrid deep learning model.

Results and discussion

The experimentation of proposed hybrid model utilizes the benchmark datasets DEAP and EEG brain wave datasets. The experiments are performed in python with essential library functions for deep learning and machine learning models. The hyperparameters details used in the experimentation is presented in Table 2. For experimentation, 80% of data are used for training and 20% of data are used for testing.

Fine-tuning the deep learning models involved in our hybrid approach is crucial for optimizing their performance on the emotion classification task. The steps and experiments undertaken to fine-tune the models are listed below.

S. no	Algorithm	Hyperparameter	value/Type
1	DenseNet 121	Learning rate	0.0001
2		Number of filters	12
3		Number of epochs	250
4		Activation function	ReLU
5		Optimizer	Adam
6	AlexNet	Maximum number of epochs	30
7		Convolutional layers	ReLU activation function
8		Number of iterations	5580
9		Fully Connected layers	SoftMax activation function

Table 2. Hyperparameters details.

Objective	Setup	Results
To determine the optimal learning rate for fine-tuning.	Tested learning rates of 0.001, 0.0001, and 0.00001.	A learning rate of 0.0001 provided the best balance between convergence speed and model accuracy.

Table 3. Experiment 1: learning Rate adjustment.

Objective	Setup	Results
To find the optimal batch size and number of epochs.	Tested batch sizes of 16, 32, and 64; epochs varied from 20 to 100.	A batch size of 32 and 50 epochs resulted in the highest validation accuracy without overfitting.

Table 4. Experiment 2: batch size and epochs.

- **Pre-training and Transfer Learning:** We initially pre-trained the AlexNet and DenseNet models on a large, publicly available dataset (ImageNet) to utilize transfer learning. This approach allows the models to benefit from learned features from a vast array of images, which helps in better initial weight settings for our specific task.
- **Dataset-Specific Fine-Tuning:** After pre-training, the models were fine-tuned on our specific datasets, the DEAP and EEG Brainwave datasets. This involved the following steps:
 - **Adjusting the Learning Rate:** We started with a lower learning rate to prevent drastic changes to the pre-trained weights. For AlexNet, we used a learning rate of 0.0001, while for DenseNet, we set it at 0.0001 as well.
 - **Optimizer Selection:** We employed the Adam optimizer for both models due to its adaptive learning rate capabilities, which help in faster convergence.
 - **Epochs and Batch Size:** We experimented with different numbers of epochs and batch sizes to find the optimal settings. For AlexNet, we used 30 epochs, and for DenseNet, we used 250 epochs. The batch size was set to 32 for all experiments.
- **Hyperparameter Tuning:** Hyperparameters such as the number of filters, activation functions, and dropout rates were carefully tuned through grid search and cross-validation techniques.
 - **Grid Search:** A range of values was tested for each hyperparameter. For example, the number of filters in the convolutional layers varied from 8 to 64, and the dropout rate was varied from 0.2 to 0.5.
 - **Cross-Validation:** We used k-fold cross-validation (k=5) to evaluate different hyperparameter combinations, selecting the configuration that provided the best performance on the validation set.
- **Regularization Techniques:** To prevent overfitting, several regularization techniques were employed:
 - **Dropout Layers:** Added dropout layers with a dropout rate of 0.4 after certain dense layers to randomly drop neurons during training.
 - **L2 Regularization:** Applied L2 regularization (weight decay) to the dense layers to penalize large weights, thus encouraging simpler models.
- **Data Augmentation:** For the DEAP and EEG Brainwave datasets, data augmentation techniques were applied to artificially increase the size and diversity of the training data. This included adding noise, scaling, and temporal shifting of the EEG signals. Table 3 give the experiment 1 with learning rate adjustment. Table 4 provides the experiment 2 with batch size epochs. Table 5 explains about the experiment 3 for regularization techniques. Table 6 shows the experiment 4 for data augmentation. Table 7 summarizes the data usage details of the proposed work. Table 8 explains the number of futures extracted and Table 9 illustrates the combined futures of extracted images.

Through these detailed fine-tuning processes and experiments, the final hybrid model combining AlexNet and DenseNet, followed by PCA and classification with a multi-class SVM, achieved superior performance in emotion classification tasks.

The training and assessment of our hybrid deep learning model, combining AlexNet, DenseNet, PCA for dimensionality reduction, and a multi-class SVM for classification, required substantial computational resources and time. Using an Intel Core i7 processor, 32GB RAM, and an NVIDIA GeForce RTX 3080 GPU, the pre-

Objective	Setup	Results
To prevent overfitting.	Tested dropout rates of 0.2, 0.3, 0.4, and 0.5; applied L2 regularization with λ values of 0.01, 0.001, and 0.0001.	A dropout rate of 0.4 and L2 regularization with $\lambda = 0.001$ provided the best validation performance.

Table 5. Experiment 3: regularization techniques.

Objective	Setup	Results
To enhance model robustness.	Applied noise addition, scaling, and temporal shifting to the training data.	Applied noise addition, scaling, and temporal shifting to the training data.

Table 6. Experiment 4: data augmentation.

Dataset	Total data points/Trials	Training set (80%)	Testing set (20%)
DEAP	1,280 trials	1,024 trials	256 trials
EEG brainwave	324,000 data points	259,200 data points	64,800 data points

Table 7. Data usage details.

Model	Number of features extracted	Layer Extracted from
AlexNet	4096	Fully connected layer fc7
DenseNet-121	1024	Global average pooling layer

Table 8. Total number of features extracted.

Model	Number of features extracted
AlexNet	4096
DenseNet-121	1024
Total features	5120

Table 9. Combined features.

training on the ImageNet dataset took approximately 3 h for AlexNet and 5 h for DenseNet. Fine-tuning on the DEAP dataset required about 2 h for AlexNet and 3 h for DenseNet, while the EEG Brainwave dataset fine-tuning took around 1.5 h for AlexNet and 2.5 h for DenseNet. Feature extraction and fusion from both models took approximately 1 h, with PCA dimensionality reduction taking an additional 30 min. Finally, training the multi-class SVM on the reduced feature set required around 45 min. The assessment phase, involving evaluation on the test set, was efficiently completed due to the reduced dimensionality of the features.

Performance metrics of proposed model

In our study, comparing the parameters of the models involved several key aspects, including the number of parameters, computational complexity, and performance metrics such as accuracy, precision, recall, F1-score, specificity, and Mathew correlation coefficient. AlexNet, with its relatively straightforward architecture, contains approximately 62 million parameters. In contrast, DenseNet-121 features about 8 million parameters, utilizing dense connections to improve feature reuse and reduce the total parameter count. The multi-class SVM's parameters depend on the support vectors and the kernel used, typically resulting in fewer parameters than deep learning models, but varying based on the dataset and feature space dimensionality. We also assessed the

S. no.	Metrics	Labels	
		Valence	Arousal
1.	Accuracy	95.54	97.26
2.	Precision	94.28	96.11
3.	F1-score	93.64	95.48
4.	Recall	93.66	93.99
5.	Specificity	92.48	93.42
6.	Mathews correlation coefficient	94.74	95.89

Table 10. Proposed model performance analysis for DEAP dataset.

S. no.	Metrics	Labels		
		Positive	Negative	Neutral
1.	Accuracy	98.78	98.48	98.00
2.	Precision	98.26	97.26	97.26
3.	Recall	97.11	97.55	97.99
4.	Specificity	98.48	98.41	98.42
5.	F1-score	97.68	97.40	97.62
6.	Mathews correlation coefficient	97.21	96.48	97.26

Table 11. Proposed model performance analysis for EEG brainwave dataset.

computational complexity by measuring training and inference times, as well as the computational resources required, such as GPU usage. All models were evaluated on a high-performance PC equipped with an Intel Core i7 processor, 32GB of RAM, and an NVIDIA GeForce RTX 3080 GPU. Performance metrics played a crucial role in our comparison, with each model being evaluated for accuracy, precision, recall, F1-score, specificity, and Mathew correlation coefficient. By analyzing these metrics, we could determine the efficacy and efficiency of each model, highlighting the strengths and weaknesses in terms of both computational demands and classification performance. This comprehensive comparison ensured that the final hybrid model, combining AlexNet and DenseNet with PCA and multi-class SVM, was optimized for both accuracy and computational efficiency in emotion classification tasks. While using DEAP dataset, the emotions are categorized as valence and arousal and the metrics are measured for each category. Table 10 depicts the performance metrics of proposed model for DEAP dataset.

Similarly for EEG brainwave dataset the metrics, like precision, recall, f1-score, specificity, Mathew correlation coefficient and accuracy are calculated. The emotions are categorized as negative, positive, and neutral. Table 11 depicts the proposed model performances for EEG brain wave dataset.

Comparative analysis

To compare the proposed model to validate the better performance average value of performance metrics are considered. For DEAP dataset, existing works like Support Vector Machine (SVM)⁴², Bidirectional Long Short-Term Memory (LSTM) network⁴³, Convolutional Neural Network (CNN)⁴⁴, Artificial Neural network (ANN)⁴⁵, and Hybrid CNN⁴⁶ are considered. The hybrid CNN which is used to compare is our own methodology in which the traditional CNN architecture is modified with support vector machine to attain improved classification performances. In the comparative analysis hybrid CNN model is mentioned as CNN + SVM for our convenience.

The precision and recall metrics attained by the proposed model for DEAP dataset is comparatively depicted in Fig. 4. The metrics is calculated for both valence and arousal categories and compared with existing methods. It is visible from the results, that maximum precision and recall metrics is attained by the proposed model for both categories compared to existing methods. The proposed model maximum precision value attained for valence and arousal category is 94.28% and 96.11%. Similarly, the proposed model maximum recall value attained for valence and arousal category is 93.66% and 93.99% respectively.

Figure 5 comparatively depicts the specificity and f1-score proposed model and existing works. The attained maximum f1-score is 93.64% and 95.48% for valence and arousal categories. Similarly, the proposed model maximum specificity score attained for valence and arousal categories are 92.48% and 93.42%, respectively. From the results, it is apparent that the proposed model performance has improved compared to existing methods due to the optimal feature extraction and classification procedures.

The comparative analysis given in Fig. 6 depicts Mathew correlation coefficient (MCC) and accuracy metrics for DEAP dataset. It is evident from the results, proposed attained maximum MCC score, and accuracy values compared to existing methods. The accuracy is greatly improved as the false positive rate is reduced in the proposed classification. The maximum MCC attained by the proposed model is 94.74% and 95.89% for valence and arousal categories respectively. Similarly, the maximum accuracy attained by the proposed model is 95.54% and 97.26% for valence and arousal category.

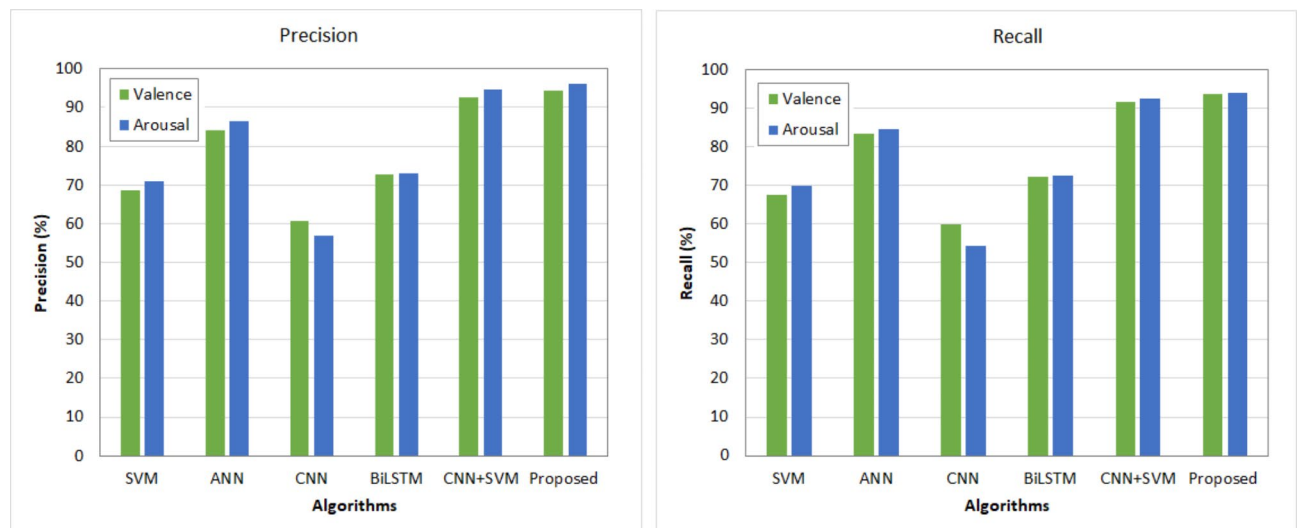


Fig. 4. Precision and recall comparative analysis.

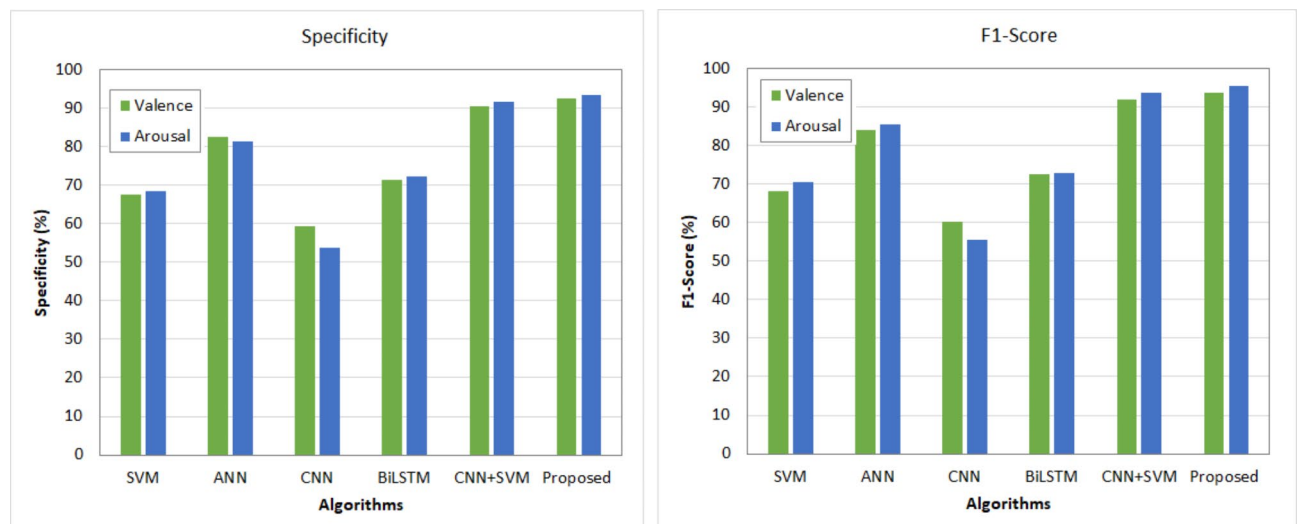


Fig. 5. Specificity and F1-score comparative analysis.

The average values attained by the proposed model is comparatively presented with existing methods for all the performance metrics in Table 12. From the results, it is clear that the proposed mode exhibited the maximum performance over all the existing methods. Further to validate the proposed model performance for EEG brainwave dataset existing methods like Long Short-Term Memory (LSTM) Network, Recurrent Neural Network (RNN), and Gated Recurrent Unit (GRU) models are considered for analysis. The results of existing works are obtained from Kalpana et al., 2022 research work. The average values of proposed model and existing methods are considered for comparative analysis.

Figure 7 depicts the precision, recall, specificity, and f1-score metrics comparative analysis for proposed model using EEG Brainwave dataset. The comparative analysis clearly depicts that the proposed model attained maximum values for all the metrics. Similarly, the accuracy obtained by the proposed model for EEG Brainwave dataset is comparatively presented in Fig. 8. The maximum accuracy attained by the proposed is 98.42%, which is much better than the existing methods.

The overall comparative analysis of proposed model and existing methods for all the metrics is presented in Table 13. From the numerical results it can be observed that the performance of the proposed model is much better than the existing methods.

From the above results, it is observed that the proposed hybrid model can effectively categorize the emotions by extracting the optimal features from the input. Using this hybrid approach, emotions and their related stress factors can be effectively classified.

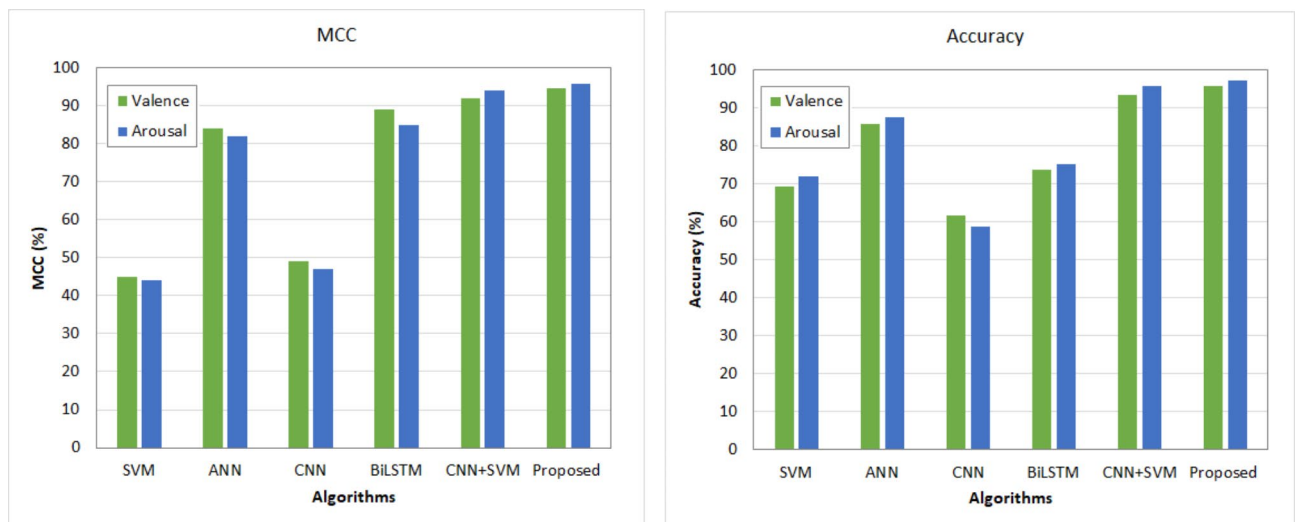


Fig. 6. MCC and accuracy comparative analysis.

S. no	Methods	Precision	Recall	F1-score	Specificity	MCC	Accuracy
1	SVM	69.82	68.71	69.26	68.01	44.50	70.55
2	ANN	85.37	84.12	84.74	82.08	83.00	86.48
3	CNN	58.71	57.05	57.87	56.53	48.00	60.00
4	BiLSTM	72.90	72.44	72.67	71.82	87.00	74.25
5	CNN + SVM	93.61	92.13	92.87	91.12	93.00	94.57
6	Proposed	95.20	93.83	94.56	92.95	95.32	96.40

Table 12. Performance comparative analysis with existing methods for DEAP dataset.

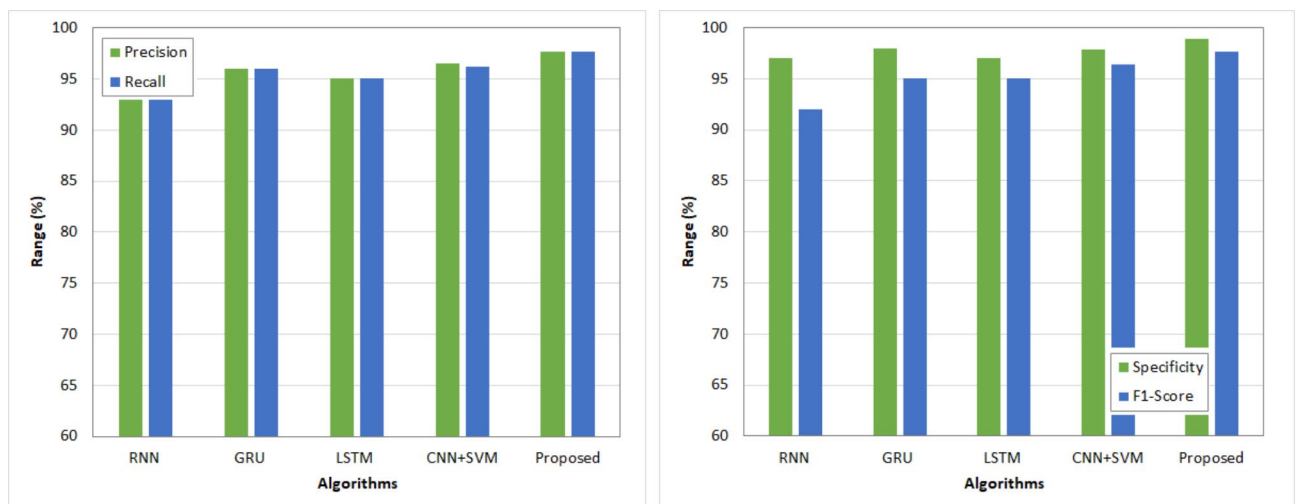


Fig. 7. Performance metrics comparative analysis.

The proposed hybrid deep learning model addresses several key limitations of traditional emotion recognition methods. Traditional machine learning models often struggle with manual feature extraction, which can be time-consuming and may fail to capture complex EEG patterns. The hybrid model uses AlexNet and DenseNet-121 for automated and efficient feature extraction, ensuring a comprehensive capture of data features. Additionally, traditional models can suffer from high dimensionality and overfitting, which the proposed model mitigates through Principal Component Analysis (PCA) for dimensionality reduction. Finally, the use of a multi-class SVM enhances classification accuracy and robustness, overcoming the limitations of simpler classifiers.

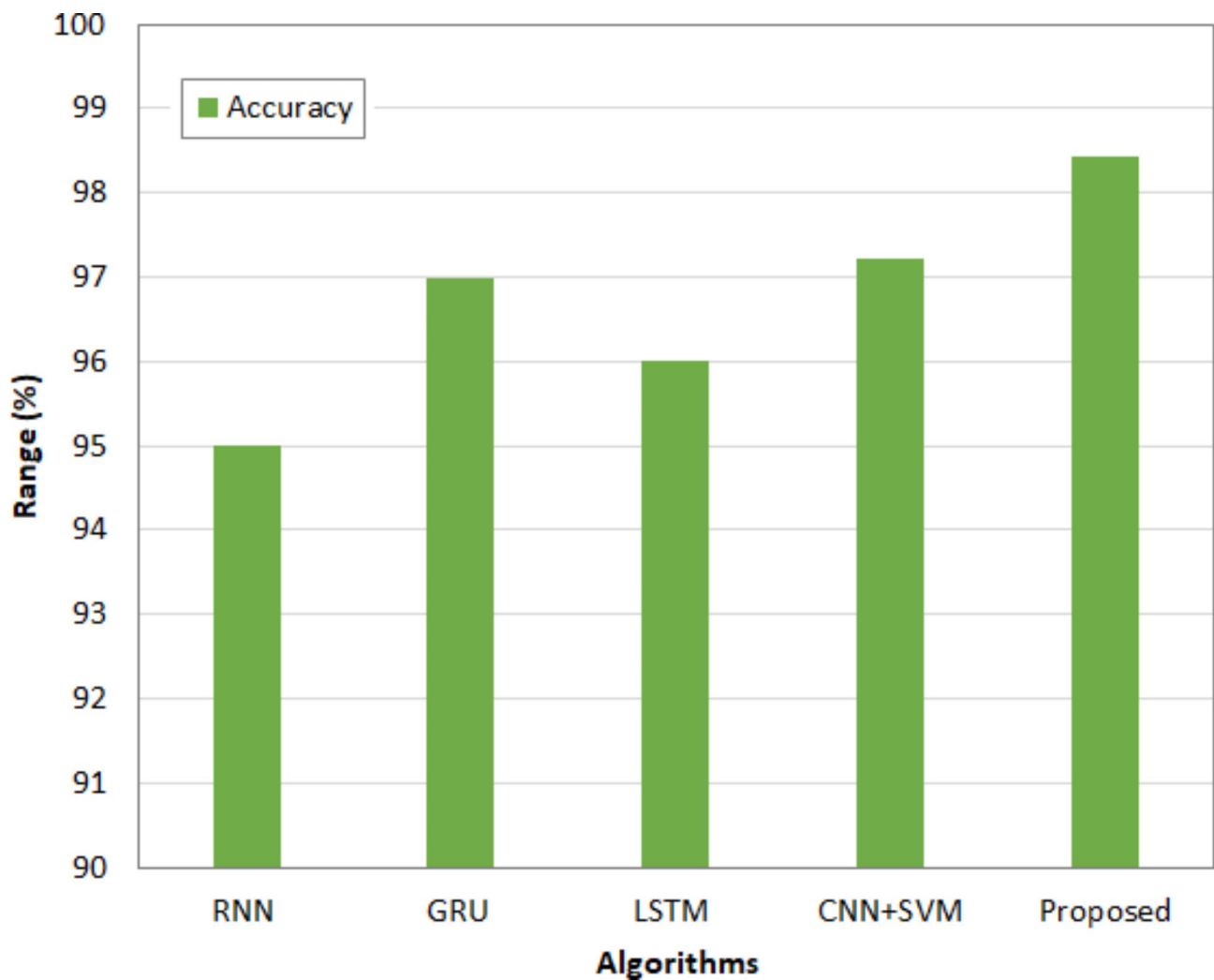


Fig. 8. Accuracy comparative analysis.

S. no	Metrics/Methods	RNN	GRU	LSTM	CNN + SVM	Proposed
1	Accuracy	95	97	96	97	98.42
2	Precision	93	96	95	96	97.59
3	Recall	93	96	95	96	97.55
4	Specificity	97	98	97	98	98.44
5	F1-score	92	95	95	96	97.57

Table 13. Performance comparative analysis with existing methods for EEG Brainwave dataset.

Conclusion

This research work presents a hybrid deep learning model for emotion classification using convolutional neural network variants like AlexNet and DenseNet. The essential data for emotion classification is processed by deep learning algorithms to extract the optimal features. The extracted features are concatenated, and their dimensionality are reduced using principal component analysis. Finally, the features are classified using multiclass support vector machine to attain improved classification performances. Experimentations using benchmark datasets DEAP and EEG Brainwave emotion dataset, validated the proposed model performances in terms of precision, recall, f1-score, specificity, Mathew correlation coefficient and accuracy. Experimental evaluation validates the superior performance of proposed model over existing machine learning and deep learning models. Future improvements for the proposed hybrid deep learning model could significantly enhance its performance and applicability. One potential direction is the incorporation of additional physiological signals such as ECG, GSR, and EMG, which can provide a more comprehensive understanding of emotional states through multimodal data integration. Advanced data augmentation techniques, including synthetic data

generation and signal transformation, can further improve the model's generalizability. Developing a real-time emotion detection system optimized for faster inference and deployment on mobile or edge devices could expand practical applications in human-computer interaction and mental health monitoring. Applying transfer learning and domain adaptation techniques can enhance the model's adaptability to different datasets or contexts, reducing the need for extensive retraining. Finally, incorporating advanced optimization algorithms like genetic algorithms, particle swarm optimization, or Bayesian optimization can fine-tune the model parameters for better classification performance and efficiency.

Data availability

The data used to support the findings of this study are included within this article.

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Author contributions

Sivasankaran Pichandi contributes to the concepts, data collection, and first draft of this article. Gomathy Balasubramanian was supported in simulation, final drafting, and overall verification. Venkatesh Chakrapani contributed to the literature review and final drafting of this article.

Declarations

Competing interests

The authors declare no competing interests.

Additional information

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